

MUHAMMED FAYIS P

Full-Stack Software Engineer · IoT & Embedded Systems · AI Integration

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PROFESSIONAL SUMMARY

Full-stack software engineer with a track record of shipping 10+ web and 7+ mobile applications across the UAE and internationally. Experienced in building production-grade backends, cross-platform mobile apps, and integrating AI/ML capabilities (OCR, LLM, Computer Vision) into real-world products. Adds a rare edge with hands-on IoT expertise — MQTT, Modbus, KNX (globally certified) — enabling end-to-end delivery from cloud to device. Based in Abu Dhabi, immediately available.

CORE SKILLS

Python · JavaScript · Java · C / C++ · Bash

Django · React JS · React Native · Node.js · HTML / CSS / Bootstrap

REST APIs · SOAP API · TCP Integration · UDP Integration · API Design & Development · Third-Party Integrations · SDK Integration · Firebase · AWS EC2

MQTT · Modbus · KNX · Zigbee · BLE · Wi-Fi · Embedded Systems · IoT Protocols

OCR Training & Integration · OpenCV · NumPy · PyTorch · TensorFlow · Redis · LLM Integration · Computer Vision · Multithreading

Linux · OpenVPN · WireGuard · OpenWRT · Postman · Cloud Hosting · Mobile & Web App Deployment

EXPERIENCE

IoT Software Engineer

2024 – Present

Sonic Technologies · Abu Dhabi, UAE

- Engineered 10+ web and 7+ mobile applications — spanning IoT dashboards, industrial monitoring, smart home control, and AI-powered platforms
- Built and deployed IoT communication layers using MQTT and Modbus protocols, bridging cloud backends with physical field devices
- Developed an OCR platform (Python + Android) for a UAE semi-government client achieving 100% recognition accuracy in production
- Built a real-time methane gas detection app (React Native + MQTT + Node.js + Firebase) for an international industrial client
- Customized embedded system firmware from scratch — independently researched, cutting hardware costs by 80%
- Integrated KNX, Zigbee, BLE, Wi-Fi, Alexa, and IP cameras into unified smart building platforms for high-rise deployments
- Configured on-premise VPN environments (OpenVPN, WireGuard) for secure remote device access and maintenance

Freelance Full-Stack Developer

2023

Independent · India / UAE

- Delivered full-stack web and mobile projects across diverse domains — e-commerce, automation, and document processing
- Built a voice-controlled car using Arduino, C programming, and sensors — integrating speech recognition with real-time hardware control

NOTABLE PROJECTS

OCR Vision Platform — Semi-Government Client, UAE

- Python + Android app processing live RTSP camera streams with custom-trained OCR (OpenCV / NumPy) — 100% accuracy in Python, 99% in Android

Realtime Industrial Gas Detection & Monitoring App

- React Native + Node.js + MQTT; Firebase push alarms; VPN-secured remote gateway access — full-stack, production deployed for international client

Smart Home Platform — High-Rise Building (KNX + IoT)

- ETS-programmed KNX system + custom mobile app; unified control of Alexa, Sonos, IP cameras, BLE, and Zigbee devices

Embedded Firmware Customization

- Reverse-engineered and modified vendor firmware to meet project specs — 80% cost reduction, fully documented, first-of-its-kind for the vendor

CERTIFICATIONS

KNX Global Certified Partner · KNX Association · Partner ID: 134980

KNX eCampus · KNX Association · Credential ID: knx706469

Career Essentials in Software Development · Microsoft & LinkedIn

Google Crash Course on Python · Google · ID: UYHBOODTBWX0

EDUCATION

B.Tech — Information Technology Engineering · APJ Abdul Kalam Technological University

ADDITIONAL

UAE Driving License (Valid) · Languages: English, Arabic, Hindi, Malayalam, Tamil